

# A Practical Guide to Image super-resolution

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TOPIC -- IMAGE SUPER-RESOLUTION

-- 01 --

**Multiplexing spatial-frequency bands** An image is formed using the normal passband of the optical device. Then, some known light structure (for example, a set of light fringes) is superimposed on the target. The image now contains components resulting from the combination of the target and the superimposed light structure, e.g. moiré fringes, and carries information about target detail which simple unstructured illumination does not. The “superresolved” components, however, need disentangling to be revealed. For an example, see structured illumination (figure to left).

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**Sub-pixel image localization** The location of a single source can be determined by computing the “center of gravity” (centroid) of the light distribution extending over several adjacent pixels (see figure on the left). Provided that there is enough light, this can be achieved with arbitrary precision, very much better than pixel width of the detecting apparatus and the resolution limit for the decision of whether the source is single or double. This technique, which requires the presupposition that all the light comes from a single source, is at the basis of what has become known as super-resolution microscopy, e.g. stochastic optical reconstruction microscopy (STORM), where fluorescent probes attached to molecules give nanoscale distance information. It is also the mechanism underlying visual hyperacuity.

-- 03 --

**Some object features, though beyond the diffraction limit, may be known to be associated with other object features that are within the limits and hence contained in the image.** Then conclusions can be drawn, using statistical methods, from the available image data about the presence of the full object. The classical example is Toraldo di Francia's proposition of judging whether an image is that of a single or double star by determining whether its width exceeds the spread from a single star. This can be achieved at separations well below the classical resolution bounds, and requires the prior limitation to the choice “single or double?” The approach can take the form of extrapolating the image in the frequency domain, by assuming that the object is an analytic function, and that we can exactly know the function values in some interval. This method is

severely limited by the ever-present noise in digital imaging systems, but it can work for radar, astronomy, microscopy or magnetic resonance imaging. More recently, a fast single image super-resolution algorithm based on a closed-form solution to  $\ell_2 - \ell_2$  problems has been proposed and demonstrated to accelerate most of the existing Bayesian super-resolution methods significantly.

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Multiple parameter use within traditional diffraction limit If a target has no special polarization or wavelength properties, two polarization states or non-overlapping wavelength regions can be used to encode target details, one in a spatial-frequency band inside the cut-off limit the other beyond it. Both would use normal passband transmission but are then separately decoded to reconstitute target structure with extended resolution.

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Super-resolution imaging (SR) is a class of techniques that improve the resolution of an imaging system. In optical SR the diffraction limit of systems is transcended, while in geometrical SR the resolution of digital imaging sensors is enhanced. In some radar and sonar imaging applications (e.g. magnetic resonance imaging (MRI), high-resolution computed tomography), subspace decomposition-based methods (e.g. MUSIC) and compressed sensing-based algorithms (e.g., SAMV) are employed to achieve SR over standard periodogram algorithm. Super-resolution imaging techniques are used in general image processing and in super-resolution microscopy.

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